Phase 1 Report: Problem Understanding, Data Quality Audit, and Exploratory Data Analysis

Predicting Hourly Taxi Surplus and Shortage Across NYC Zones Using Net-Flow (Dropoffs − Pickups) (Path A)

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1 Problem Formulation

In a busy city like New York, taxi companies constantly struggle to keep their fleet balanced across different neighborhoods. Most simple analyses only count **Pickups** (how many passengers request a ride). However, every taxi trip affects two locations:

1. When a passenger is **picked up**, that origin zone **loses 1 available taxi** (Outflow).
2. When a passenger is **dropped off**, that destination zone **gains 1 empty, available taxi** (Inflow).

Because pickups and dropoffs do not happen at the same rate in the same place, two major operational problems occur every day:

- **Taxi Surplus (Dropoffs > Pickups):** Too many taxis drop off passengers in one area (such as Midtown Manhattan at 8:30 AM) and cannot find new passengers. Drivers waste fuel driving around empty, earn less money, and create extra traffic congestion.
- **Taxi Shortage (Pickups > Dropoffs):** More people need a taxi than there are taxis arriving (such as residential neighborhoods at 8:00 AM, or office areas at 6:00 PM when workers go home). Passengers face long wait times or cancel their rides.

Decision Supported by This Analysis: Our project supports a real operational decision: **taxi repositioning**. By predicting the hourly **Net Flow (Net Flow = Dropoffs − Pickups)** for each zone, dispatchers can see ahead of time which zones will have extra empty taxis (+) and guide those drivers to nearby zones that are about to face a shortage (−).

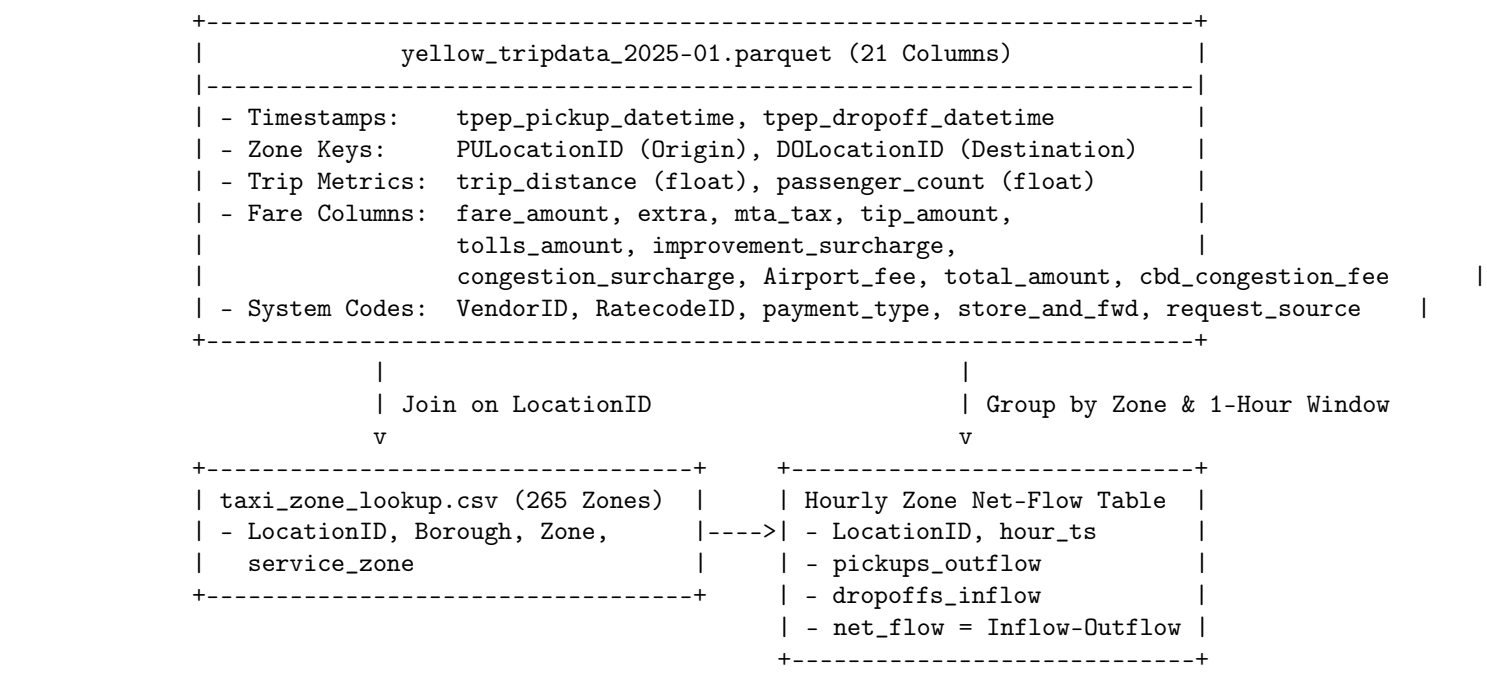
2 Understanding the Data Source

- **Data Source:** Official public trip records published by the **New York City Taxi and Limousine Commission (NYC TLC)**.
- **Collection Method:** Data is recorded automatically by electronic taximeters (TPEP systems) installed in NYC Yellow Taxis, operated by two technology vendors (`VendorID = 1`: Creative Mobile Technologies, and `VendorID = 2`: Curb Mobility).
- **Observational vs. Controlled:** This is **observational data** recorded from real city trips, not a controlled experiment.
- **Time Period and Selected Subset:** We analyze all Yellow Taxi trips from **January 2025** (`yellow_tripdata_2025-0` paired with the official neighborhood lookup table (`taxi_zone_lookup.csv`, which maps 265 Location IDs to borough and zone names).

- **Unit of Observation vs. Unit of Prediction:**
  - Unit of Observation (Raw Data): A single completed taxi ride.
  - Unit of Prediction (Grouped Table): One specific **Taxi Zone during a 1-hour window** (`LocationID`, `hour_ts`).
- **Main Limitations of the Data:**
  1. Unrecorded Unmet Demand: Taximeters only log rides that actually happened. If a passenger waited on the street in a zone with zero taxis and gave up, that missed demand is not recorded.
  2. Sensor and Input Glitches: GPS dropouts can record 0 miles for a paid trip, and drivers often skip typing the passenger count.
  3. Geographic Bias: Yellow taxis mainly serve Manhattan and the airports (JFK and LaGuardia). Outer boroughs rely more on Uber/Lyft and Green Taxis.

3 Data Structure Audit

The raw dataset contains **21 columns** covering timestamps, location IDs, trip metrics, fare breakdowns, and vendor flags, which we join with `taxi_zone_lookup.csv` (265 rows, 4 columns).



4 Data Quality Audit

We audited the January 2025 dataset for missing values, duplicate records, and physically impossible numbers.Initial decisions and The counts of each issue are plotted in Figure 1 and summarized in Table 1.

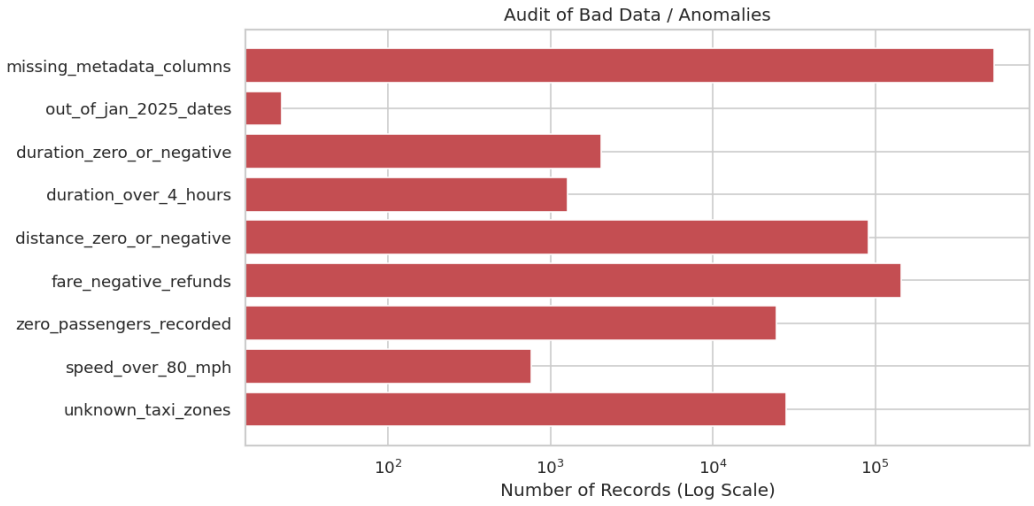


Figure 1: Audit of Data Quality Issues and Physical Anomalies (Log Scale).

4.1 Data Cleaning

As you eill see in the results of the notebook there is no exact duplicates in the dataset.
16 out of 21 columns have zero missing values. However, 5 metadata columns (`passenger_count`, `RatecodeID`, `store_and_fwd_flag`, `congestion_surcharge`, and `Airport_fee`) are missing in **the exact same 540,149 rows**.

- **Reason:** These values are not missing at random. They belong to app-booked rides (E-Hail / Flex-Fare) where the fare is calculated through a mobile app rather than typed into the traditional taximeter.
- **Initial Decision:** Because the pickup/dropoff times and zone IDs are completely valid on these 540,149 rows, deleting them would erase over half a million real taxi rides and ruin our zone flow counts. Therefore, we keep them in our hourly Net-Flow table and create an `is_ehail` flag for trip-level analysis.

Table 1: Quantified Data Quality Anomalies in January 2025 and Initial Decisions

Issue / Anomaly Type	Condition Checked	Row Count	Initial Handling Decision
Missing Metadata Block	`passenger_count` is Null	540,149	Keep for flow counts; add `is_ehail` flag
Negative Fare (Refunds)	`fare_amount < 0`	144,118	remove
Zero or Negative Distance	`trip_distance <= 0`	90,893	Remove
Unknown Taxi Zones	`LocationID` in {264, 265}	28,357	Remove
Zero Passengers Recorded	`passenger_count == 0`	24,656	Driver input error; replace with median
Zero or Negative Duration	`duration_min <= 0`	2,051	Remove
Duration Over 4 Hours	`duration_min > 240`	1,278	Remove
Speed Over 80 mph	`speed_mph > 80`	766	Remove
Wrong Dates (Not Jan 2025)	Outside `2025-01`	22	Remove

5 Univariate Exploratory Data Analysis

After removing impossible physical outliers (keeping rides between 1 and 180 minutes, 0.1 to 50 miles, $1 to $250 fare, and 1 to 70 mph speed), Figure 2 shows the distributions of the main variables, and Table 2 reports their summary statistics.

Table 2: Univariate Summary Statistics for Core Trip and Flow Variables

Variable	Mean	Std Dev	Median (50%)	Q1 (25%)	Q3 (75%)	IQR	Skewness
`trip_distance` (miles)	3.18	4.09	1.70	1.02	3.16	2.14	+3.15
`duration_min` (min)	14.71	11.29	11.72	7.37	18.35	10.98	+2.47
`fare_amount` ($ USD)	17.87	15.74	12.80	8.60	19.80	11.20	+2.96
`speed_mph` (mph)	11.36	6.36	9.64	7.42	13.04	5.61	+1.99
`net_flow` (per zone-hr)	0.00	32.99	1.00	-1.00	+4.00	5.00	-4.67

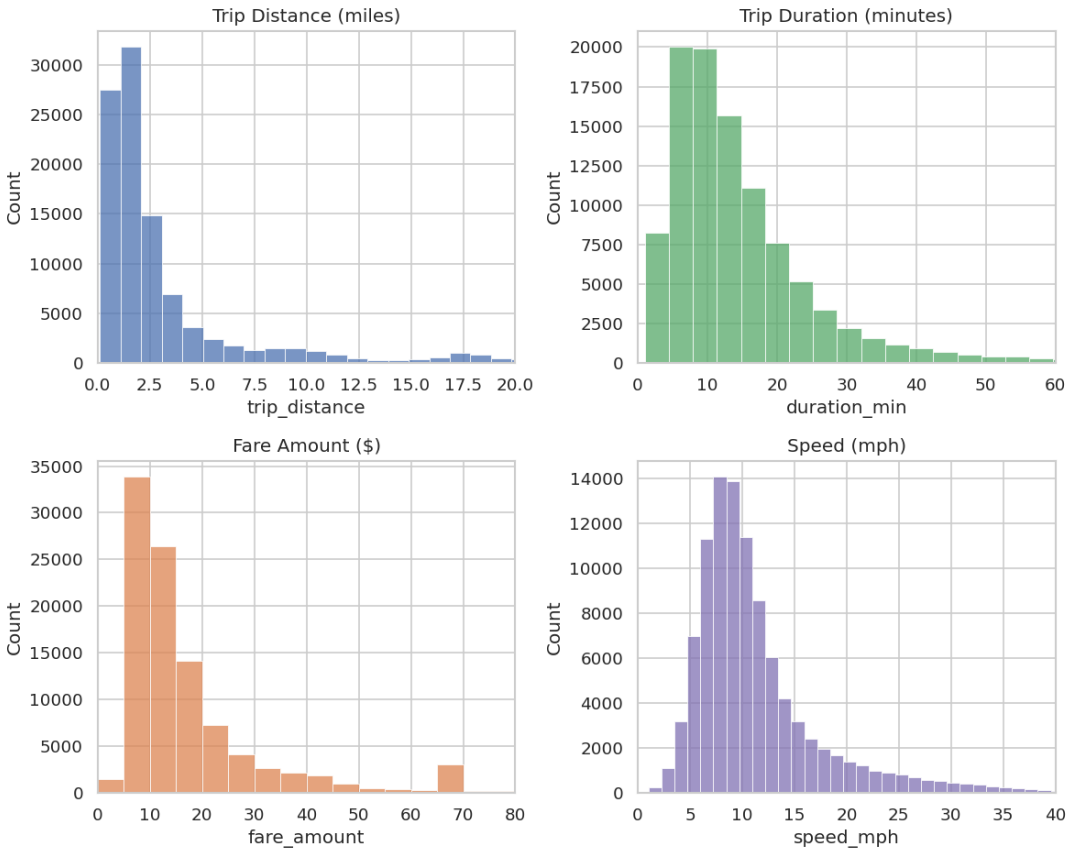


Figure 2: Univariate Distributions of Trip Distance, Duration, Fare Amount, and Speed.

Key Univariate Findings:

1. **Right-Skewed Distributions:** Most NYC taxi rides are short neighborhood trips (median distance is **1.70 miles** and median duration is **11.72 minutes**). However, long rides to JFK and LaGuardia airports (15–22 miles) create a long right tail and a small second bump around $70 in the fare chart.

2. **Slow Urban Traffic:** The median taxi speed is just **9.64 mph** due to heavy street congestion and traffic lights.

6 Auditing Trip Times

Figure 3 displays the daily trip totals across January 2025 (left) and a heatmap of trip volume by Day of Week and Hour of Day (right).

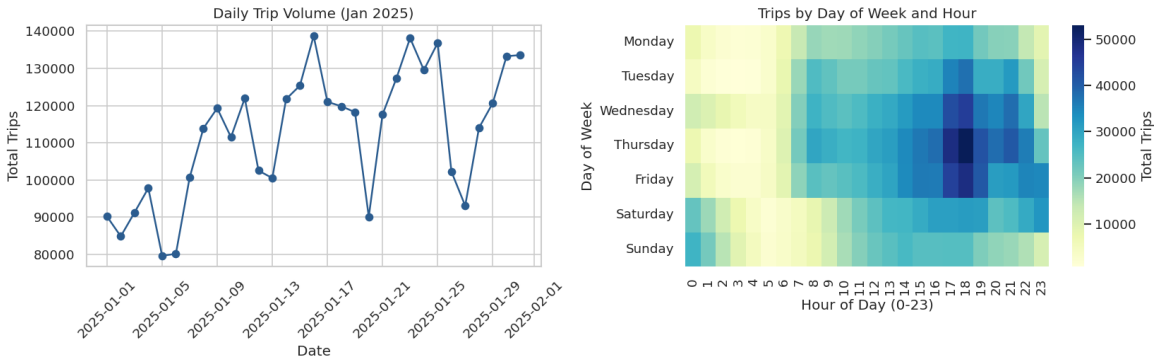


Figure 3: Daily Trip Volume by Date (Left) and Hourly Heatmap by Day of Week (Right).

Key Temporal Findings:

- **Weekday Rush Hours vs. Weekend Nightlife:** On weekdays (Monday–Friday), trips rise sharply after 6:00 AM and reach their highest peak between **17:00 and 19:00** when commuters leave work. On Saturday and Sunday mornings (6:00–10:00 AM), the city is quiet, but there is heavy demand between **00:00 and 04:00 AM** from weekend nightlife.
- **Holiday and Weather Drops:** Daily volume drops noticeably on **Jan 1 (New Year's Day)** and mid-month around **MLK Jr. Day / winter weather**, showing clear calendar effects.

7 Consistency of Trip Variables and Multivariate EDA

Figure 4 checks the relationship between Distance and Fare (left) and how average speed changes across the 24 hours of the day (right). Figure 5 shows the correlation matrix across trip variables.

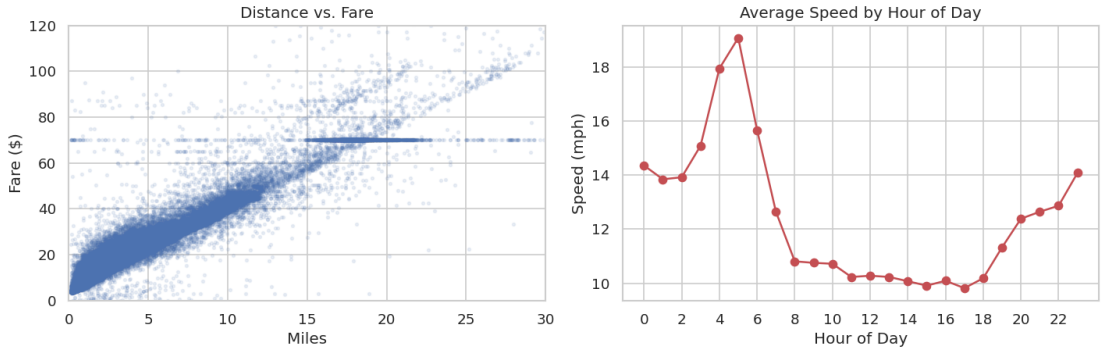


Figure 4: Distance vs. Fare Consistency (Left) and Average Speed by Hour (Right).

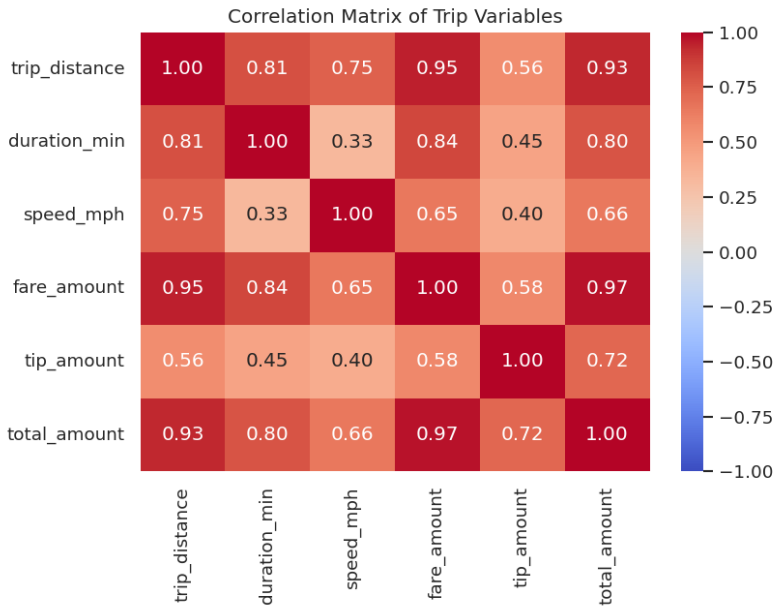


Figure 5: Multivariate Correlation Matrix of Core Trip Variables.

Key Consistency and Multivariate Findings:

1. **Why is there a flat horizontal line at $70?** Distance and fare have a very strong linear correlation ($r = 0.91$). However, the scatter plot in Figure 4 (left) shows a flat horizontal line at **exactly $70.00** across many distances. This is not an error: NYC charges a fixed **$70 flat fare** (`RatecodeID = 2`) for trips between Manhattan and JFK Airport.

2. **Time of Day Controls Speed and Duration:** As seen in Figure 4 (right), taxis travel at nearly **18 mph at 5:00 AM**, but slow down to **under 10 mph between 12:00 PM and 5:00 PM**. Because of this, the same 5-mile trip takes twice as long in the afternoon as it does early in the morning.

8 Initial Spatial View and Net-Flow Dynamics

Figure 6 shows the top 15 pickup zones and the cumulative share of pickups across all zones. Figure 7 compares the hourly **Net Flow (Dropoffs − Pickups)** between a commercial office hub (`Midtown Center`) and a residential hub (`Upper East Side`).

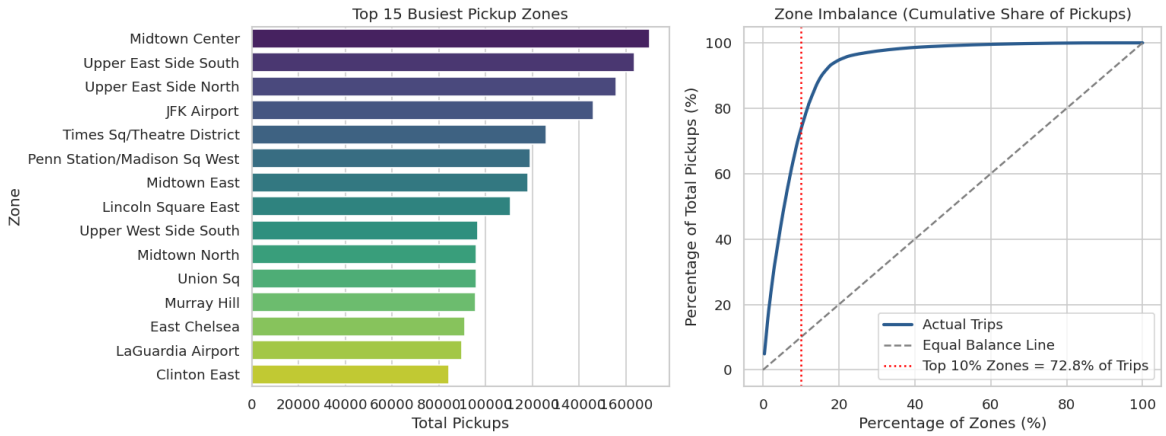


Figure 6: Top 15 Busiest Pickup Zones (Left) and Cumulative Zone Concentration Curve (Right).

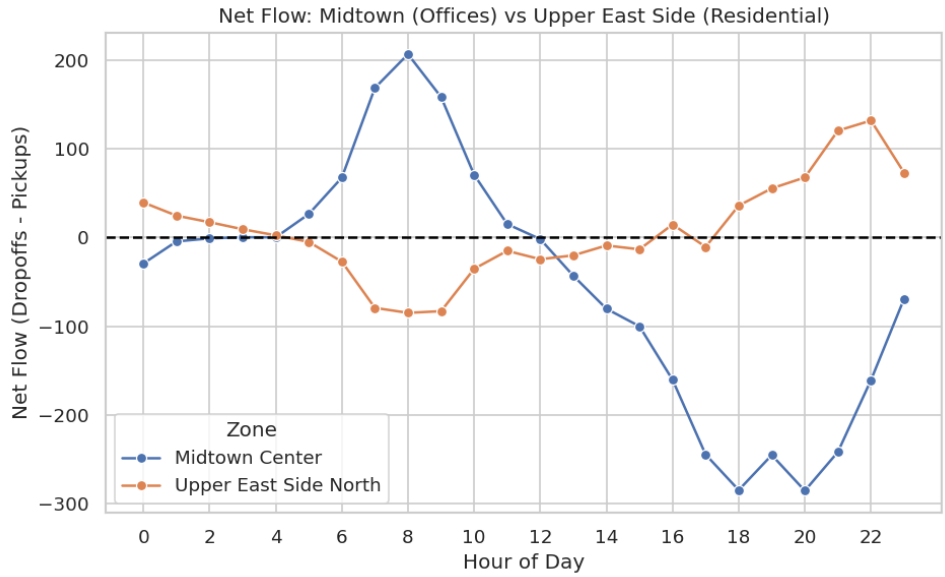


Figure 7: Hourly Net-Flow (Dropoffs − Pickups) Comparison: Office Zone vs. Residential Zone.

Key Spatial and Net-Flow Insights:

1. **Severe Spatial Imbalance:** Figure 6 (right) shows that just 10% of NYC taxi zones generate over 70% of all pickups. Most activity happens in Manhattan and at JFK/LaGuardia airports.

2. **Opposite Daily Waves of Surplus and Shortage (Figure 7):**

   - **Midtown Center (Office Zone):** Between **7:00 AM and 10:00 AM**, commuters arrive at work, creating a huge **Taxi Surplus (+150 empty taxis/hour)**. Between **5:00 PM and 10:00 PM**, workers leave for home, flipping Midtown into a severe **Taxi Shortage (−160 taxis/hour)**.
   - **Interesting Airport Rule (Newark Airport):** In our Net-Flow table, `Newark Airport` (in New Jersey) shows positive dropoffs (e.g., 13 dropoffs at 4:00 AM) but **0 pickups**. This reflects real NYC law: Yellow Taxis can drop passengers off at Newark Airport, but are not legally allowed to pick up new passengers in New Jersey.

- **Upper East Side (Residential Zone):** Shows the exact opposite pattern—a **Morning Shortage (−90 taxis/hour)** when residents leave for work, and an **Evening Surplus (+80 taxis/hour)** when they return home.

9 Data Leakage Risks and Proper Data Split

There is the potenitial of three sources of data leakage:

1. **Future Time Leakage:** Splitting time-series data randomly would let the model use 3:00 PM traffic to predict 2:00 PM traffic on the same day. **Proper Split Unit:** We use a strict chronological date split:
   - **Train Set:** Jan 1 – Jan 21, 2025
   - **Validation Set:** Jan 22 – Jan 26, 2025
   - **Test Set:** Jan 27 – Jan 31, 2025
2. **Same-Hour Dropoff Leakage:** When predicting `net_flow` for hour t (e.g., 8:00–9:00 AM), we cannot use trips that finish at 8:30 AM as input features. We only use trips completed strictly before 8:00 AM.
3. **Preprocessing Leakage:** All data preprocessing works tha might cause data leakage in next phases will be fit only on the Train set (Jan 1–21).

10 Defining a Simple Baseline

To prove that our machine learning models in next phases actually add value, we compare them against two simple baseline rules:

1. **24-Hour Naive Baseline (Yesterday Same Hour):** Predict that a zone's `net_flow` at hour t equals its `net_flow` at the exact same hour yesterday ($\hat{y}_{z,t} = y_{z,t-24}$).
2. **168-Hour Naive Baseline (Last Week Same Hour):** Predict that a zone's `net_flow` at hour t equals its `net_flow` at the exact same hour 7 days ago ($\hat{y}_{z,t} = y_{z,t-168}$).

11 Three Initial Questions and Final Selection

We evaluated three candidate questions for this dataset:

1. **Question 1 :** Can we predict the hourly **Net Flow (Dropoffs − Pickups)** in each NYC taxi zone to help dispatchers move empty taxis from surplus zones to shortage zones?
2. **Question 2 :** Can we predict how many minutes a trip will take (`duration_min`) based on origin, destination, distance, and hour of day?
3. **Question 3 :** What factors determine how much tip (`tip_amount`) a passenger gives?

Why We Chose Question 1: Question 3 is flawed because cash tips are always recorded as \$0 by the taximeter, creating a biased label. While Question 2 is useful, Question 1 is much richer because it uses both Pickup and Dropoff locations, fulfills Path A's time-series and spatial requirements naturally, allows us to cluster neighborhoods by their daily flow waves in Phase 2, and directly solves the real-world problem of empty cruising taxis.